



MANH-CUONG NGUYEN

WORK EXPERIENCE

CONDUCTED RESEARCH ON EQUIPMENT FAULT DIAGNOSIS USING VIBRATION SIGNALS AND AI-BASED METHODS (2025)

Implemented and deployed the AI model on an Edge Device for real-time fault detection

Applied AI techniques for vibration-based fault diagnosis of rotating machinery

EDUCATION

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

Bachelor at Mechatronics Engineering (*Good Graduation*)

English skills: B2

PERSONAL PROJECT

AutoDub-Lite: End-to-End AI Video Dubbing Pipeline
github.com/deth102/AutoDub_Lite

Built an end-to-end AI dubbing pipeline that automatically converts English YouTube videos into synchronized Vietnamese dubbed versions using STT, translation, and TTS.

Real-Time Fault Diagnosis on Edge Devices
github.com/deth102/Realtime_fault_diagnosis_on_edgedevice

Deployed Softmax Regression and Random Forest models on edge hardware for real-time, standalone fault diagnosis directly from raw vibration signals.

Random Forest & Logistic Regression for Fault Diagnosis
github.com/deth102/random_forest_logistic_regression

Developed a vibration-based fault diagnosis pipeline with time-frequency analysis and advanced signal preprocessing. Optimized Random Forest and Logistic Regression models for accurate fault classification.

CNN Transfer Learning for Fault Diagnosis
github.com/deth102/cnn-transferlearning

Applied CNN-based transfer learning to adapt models across operating conditions, improving diagnostic performance with limited labeled target-domain data.

Transformer-Based Fault Diagnosis
github.com/deth102/transformer_for_fault_diagnosis

Developed Conv-Transformer and Vision Transformer models for time-series and time-frequency data, achieving accurate diagnosis of complex mechanical faults.

Softmax Regression in Python & Softmax Regression in C++
github.com/deth102/softmax_regression

Implemented a softmax-based neural network from scratch without ML/DL libraries and successfully deployed it in C++, enabling low-level execution and embedded AI.

CONTACT

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BASIC INFORMATION

Birthday: December 18, 2003

Nationality: Vietnamese

Sex: Male

REASEARCH FIELDS

Signal Processing for Fault Detection

Vibration Analysis and Diagnosis

Prognostics and Health Management (PHM)

Remaining Useful Life (RUL) Prediction

Explainable AI (XAI) for Fault Diagnosis

Transfer Learning for Fault Diagnosis

TECHNICAL SKILLS

PROGRAMING LANGUAGES	SOFTWARES
Python	Autocad
C/C++	Solidworks
Matlab & Simulink	VS code
	Office